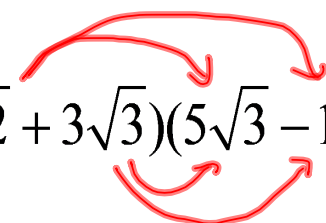


## 1.5 Solving Quadratic Equations

### *Part 1: Solve by Factoring*

**DO IT NOW!**

1. Simplify.  $(\sqrt{2} + 3\sqrt{3})(5\sqrt{3} - 10)$



$$= \sqrt{2}(5\sqrt{3}) + \sqrt{2}(-10) + 3\sqrt{3}(5\sqrt{3}) + 3\sqrt{3}(-10)$$

$$= 5\sqrt{6} - 10\sqrt{2} + 15\sqrt{9} - 30\sqrt{3}$$

$$= 5\sqrt{6} - 10\sqrt{2} - 30\sqrt{3} + 45$$

2. Simplify  $\frac{2 - \sqrt{80}}{4}$

$$= \frac{2 - \sqrt{16}(\sqrt{5})}{4}$$

$$= \frac{2 - 4\sqrt{5}}{4}$$

$$= \frac{2}{4} - \frac{\cancel{4}\sqrt{5}}{\cancel{4}}$$

$$= \frac{1}{2} - \sqrt{5}$$

3. Expand and simplify:

$$4\sqrt{10}(3 + 2\sqrt{2})$$

$$= 4\sqrt{10}(3) + 4\sqrt{10}(2\sqrt{2})$$

$$= 12\sqrt{10} + 8\sqrt{20}$$

$$= 12\sqrt{10} + 8\sqrt{4}(\sqrt{5})$$

$$= 12\sqrt{10} + 16\sqrt{5}$$

# Lesson Outline

**Section 1:** Solve a quadratic with an ' $a$ ' value of 1 or that can be factored out

**Section 2:** Solve a quadratic with an ' $a$ ' value of not 1 that can't be factored out.

\*In all cases we will start with an equation in Standard Form and we will set it equal to 0:

$$ax^2+bx+c = 0$$

**NOTE:** If it's not in standard form, you must rearrange before factoring.

## **HOW TO SOLVE QUADRATICS**

Solving a quadratic means to find the x-intercepts or roots.

**To solve a quadratic equation:**

- 1) It must be set to equal 0. Before factoring, it must be in the form  $ax^2+bx+c = 0$
- 2) Factor the left side of the equation
- 3) Set each factor to equal zero and solve for ' $x$ '.

**zero product rule:** if two factors have a product of zero; one or both of the factors must equal zero.

**Example 1:** Solve the following quadratics by factoring

a)  $y = x^2 - 15x + 56$

p: 56  
s: -15  
-8 and -7

$$y = (x-8)(x-7)$$

$$0 = (x-8)(x-7)$$

$$x-8=0 \quad \text{or} \quad x-7=0$$

$$x=8$$

$$x=7$$

When factoring  $ax^2+bx+c=0$  when 'a' is 1 or can be factored out

**Steps to follow:**

- 1) Check if there is a common factor that can be divided out
- 2) Look at the 'c' value and the 'b' value
- 3) Determine what factors multiply to give 'c' and add to give 'b'
- 4) put those factors into  $(x+r)(x+s)$  for 'r' and 's'.
- 5) make sure nothing else can be factored

b)  $y = x^2 - 5x + 6$

p: 6  
s: -5  
-2 and -3

$$0 = (x-2)(x-3)$$

$$0 = (x-2)(x-3)$$

$$x-2=0 \quad \text{or} \quad x-3=0$$

$$x=2$$

$$x=3$$

c)  $y = 2x^2 - 8x - 42$

p: -21  
s: -4  
-7 and 3

$$0 = 2(x^2 - 4x - 21)$$

$$0 = 2(x-7)(x+3)$$

$$0 = (x-7)(x+3)$$

$$x-7=0 \quad \text{or} \quad x+3=0$$

$$x=7$$

$$x=-3$$

Steps to factoring  $ax^2 + bx + c$  when 'a' cannot be factored out and is not 1.

- 1) Look to see if there is a common factor that can be divided out
- 2) Take the 'a' value and multiply it to the 'c' value
- 3) Determine what factors of THIS number add together to get the 'b' value
- 4) Break the 'b' value up into THOSE factors!
- 5) Put parenthesis around the first two variables and the last two
- 6) Factor by grouping

**Example 2:** Solve by factoring

a)  $8x^2 + 2x - 15 = 0$

P: -120  
S: 2

12 and -10

$8x^2 + 12x - 10x - 15 = 0$

(factor by grouping)

$(8x^2 + 12x) + (-10x - 15) = 0$

(common factor each group)

$4x(2x+3) - 5(2x+3) = 0$

(binomial common factor)

$(2x+3)(4x-5) = 0$

(zero product rule)

$2x+3=0$  or  $4x-5=0$

$x = -\frac{3}{2}$

$x = \frac{5}{4}$

b)  $2x^2 - 11x = -15$

p: 30  
s: -11

-6 and -5

$$2x^2 - 11x + 15 = 0$$

$$2x^2 - 6x - 5x + 15 = 0$$

(factor by grouping)

$$(2x^2 - 6x) + (-5x + 15) = 0$$

(common factor each group)

$$2x(x - 3) - 5(x - 3) = 0$$

(binomial common factor)

$$(x - 3)(2x - 5) = 0$$

(zero product rule)

$$x - 3 = 0 \quad \text{or} \quad 2x - 5 = 0$$

$$x = 3$$

$$x = \frac{5}{2}$$

**Example 3:** For the quadratic  $y = 2x^2 - 4x - 16$

a) Find the roots of the quadratic by factoring

$$0 = 2(x^2 - 2x - 8)$$

p: -8  
s: -2

-4 and 2

$$0 = x^2 - 2x - 8$$

$$0 = (x - 4)(x + 2)$$

$$x - 4 = 0 \quad \text{or} \quad x + 2 = 0$$

$$x = 4$$

$$x = -2$$

b) Find the axis of symmetry (average of x-intercepts)

$$\text{a.o.s: } x = \frac{r+s}{2} = \frac{4+(-2)}{2} = \frac{2}{2} = 1$$

c) Find the coordinates of the vertex and state if it is a max or min value

$$\begin{aligned}x_{\text{vertex}} &= 1 & y_{\text{vertex}} &= 2x^2 - 4x - 16 \\& & &= 2(1)^2 - 4(1) - 16 \\& & &= 2 - 4 - 16 \\& & &= -18\end{aligned}$$

∴ vertex is  $(1, -18)$

This is a minimum value because the parabola opens up ( $a > 0$ ).